

The development of the model of the atom

Atomic structure
Paper 1 · Combined and Separate
14 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 5 starred (★) questions first.

	QUESTION	ANSWER	✓
1	Name the first subatomic particle to be discovered.	The electron	☐
2 ★	Name the model of the atom with electrons embedded in a positively charged sphere.	The plum pudding model	☐
3	Name the model of the atom proposed after the nucleus was discovered.	The nuclear model	☐
4	Name the experiment that discovered the nucleus of an atom.	The alpha particle scattering experiment	☐
5	State what the alpha particle scattering experiment showed about the nucleus.	The mass of the atom is concentrated in a very small, positively charged nucleus.	☐
6 ★	Explain how the alpha particle scattering experiment showed that the nucleus is very small.	Most alpha particles passed straight through; only a very few were scattered backwards.	☐
7 ★	Explain how the alpha particle scattering experiment showed that the nucleus is positively charged.	Some positive alpha particles were deflected or scattered backwards, repelled by the nucleus.	☐
8	Name the model of the atom that came after the nuclear model.	The Bohr model (Bohr adapted the nuclear model)	☐
9 ★	Describe the Bohr model of the atom.	A small positive nucleus with electrons orbiting at specific distances.	☐
10	What were atoms thought to be before the electron was discovered?	Tiny spheres that could not be divided.	☐
11	Most alpha particles passed straight through the gold foil. State what this showed.	Most of the atom is empty space.	☐

	QUESTION	ANSWER	✓
12	Explain why the alpha particle scattering experiment led to the plum pudding model being replaced.	It could not explain why some alpha particles were deflected through large angles.	☐
13 ★	Name the scientist who provided the evidence for the neutron.	James Chadwick	☐
14	Name the scientist who suggested that electrons orbit at specific distances.	Niels Bohr	☐